

# 软件无线电开发板

## **SDR-Z103**

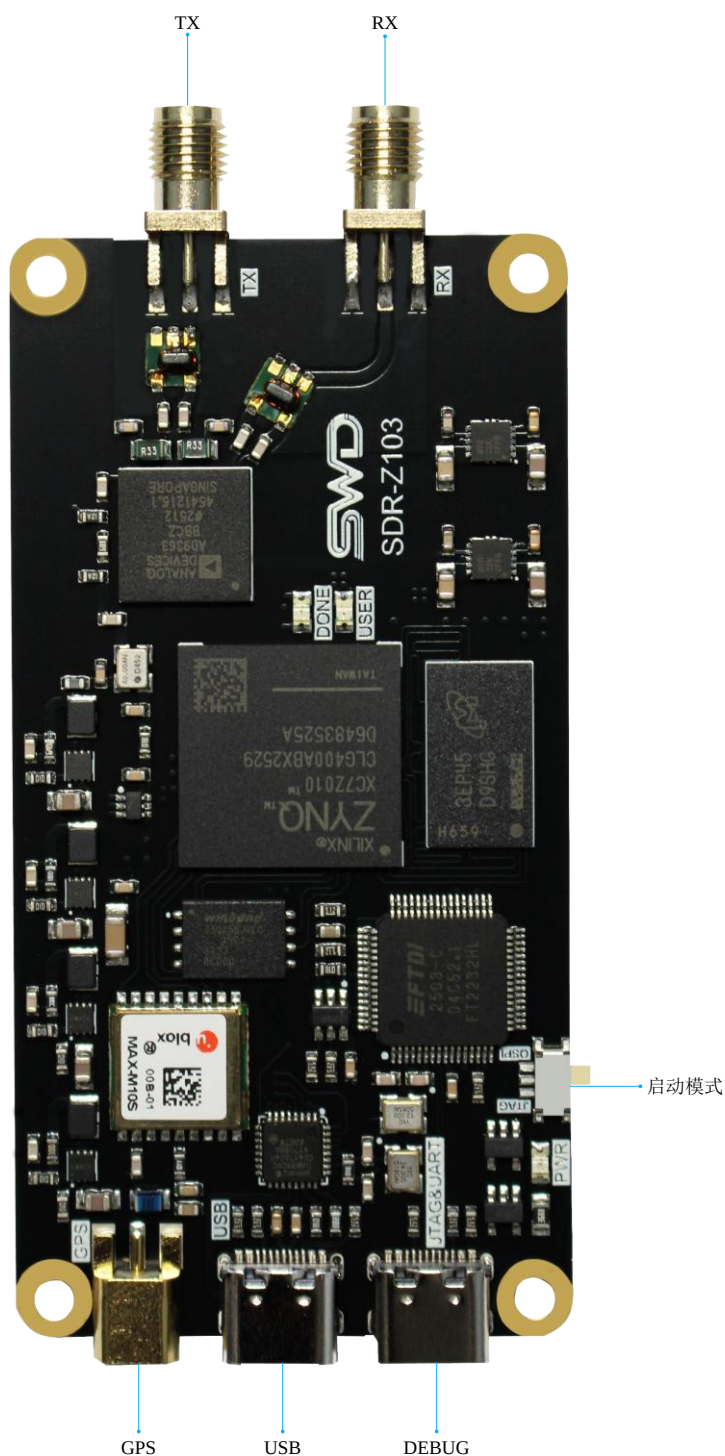
## 固件烧录指南

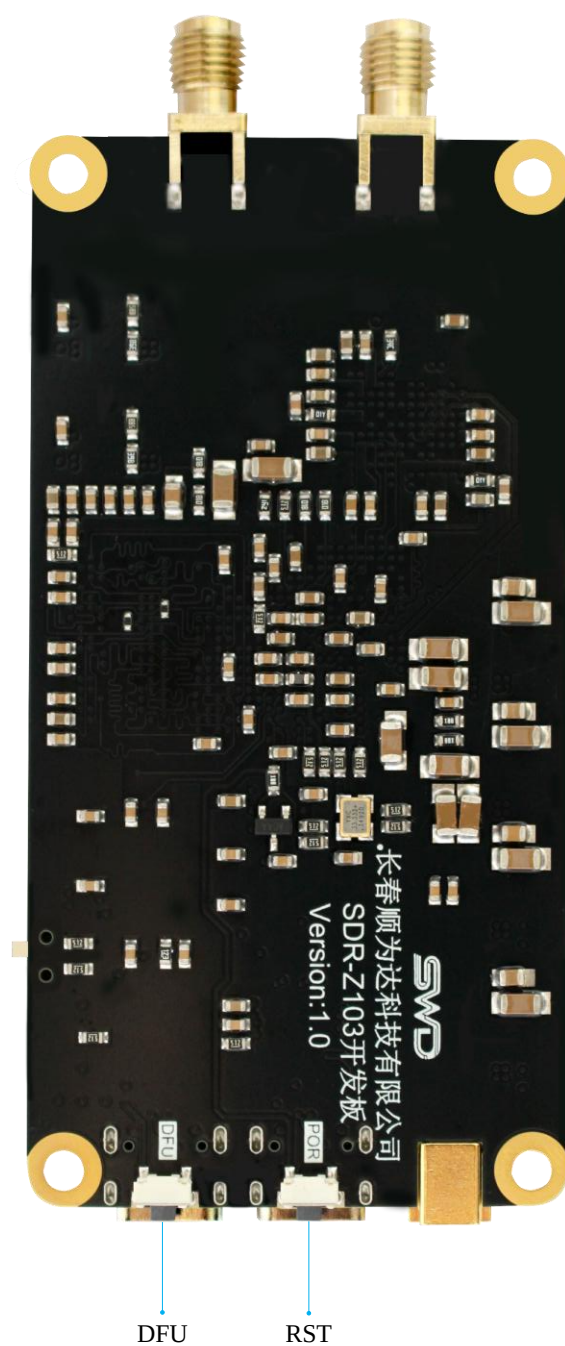
长春顺为达科技有限公司

## 前言

感谢您使用顺为达出品的 SDR-Z103 开发板，本文档旨在帮助用户能够快速完成固件烧录。

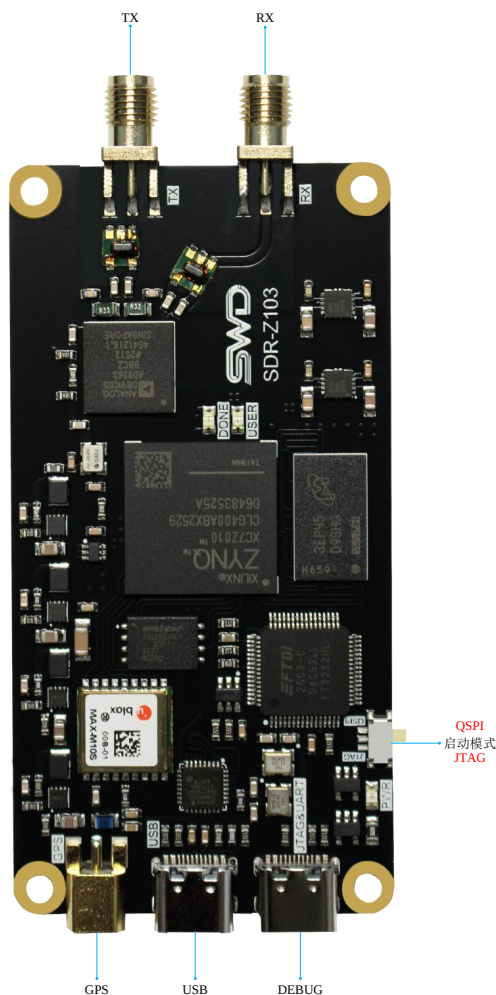
本产品所有资料不讨论 FPGA、ARM、SDR 等基础学习相关内容，提供的例程和文档仅供参考，请具备相关能力的用户自行研究与开发。



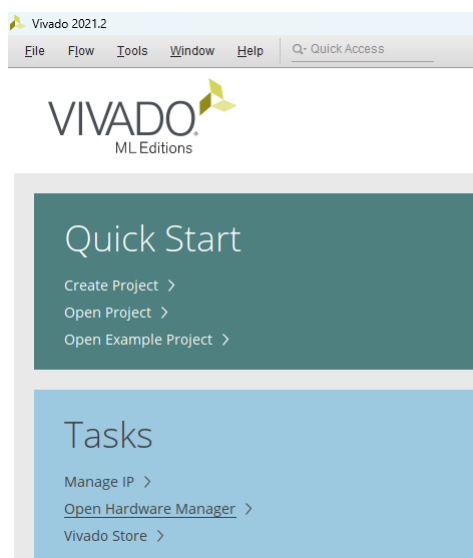


固件烧录会用到 Xilinx Vivado 和 Vitis 或 SDK 软件，请自行安装。

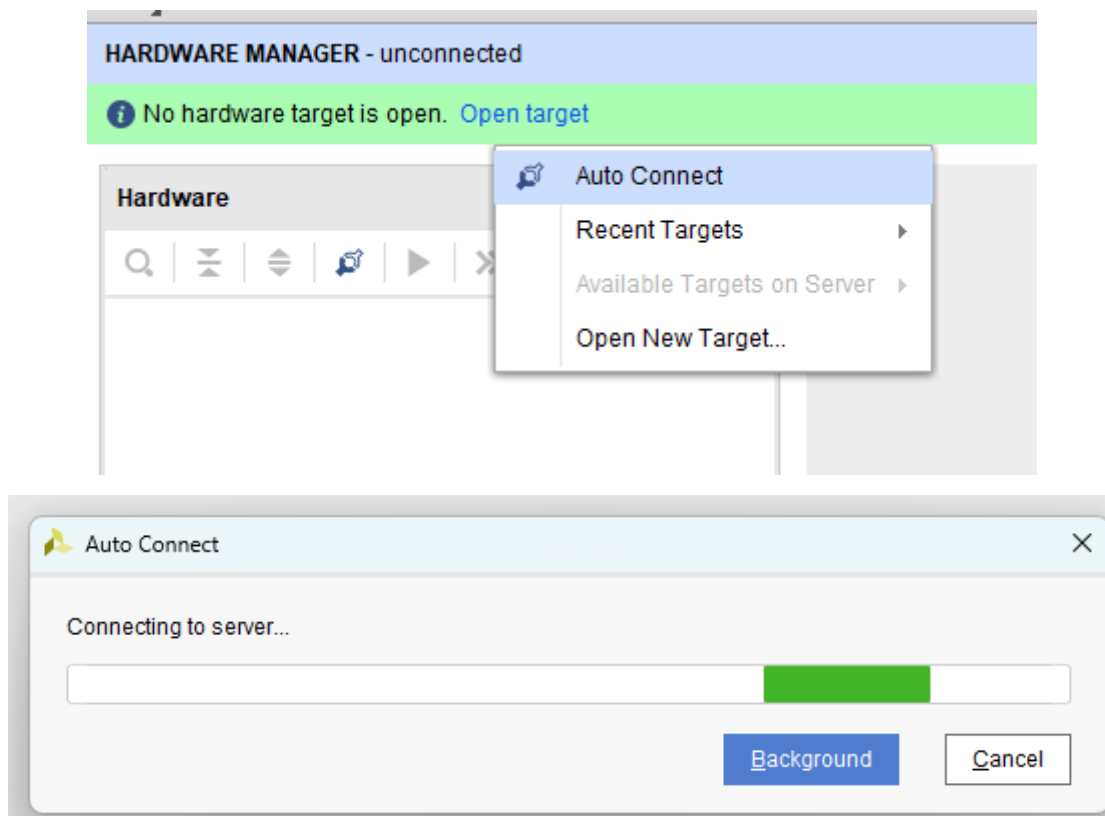
将板卡启动模式设置为 JTAG，将 DEBUG 口线缆插入到计算机 USB 口。  
可先打开 Vivado 软件，连接硬件，查看硬件在线状态。



- 打开 Vivado，点击“Open Hardware Manager”。



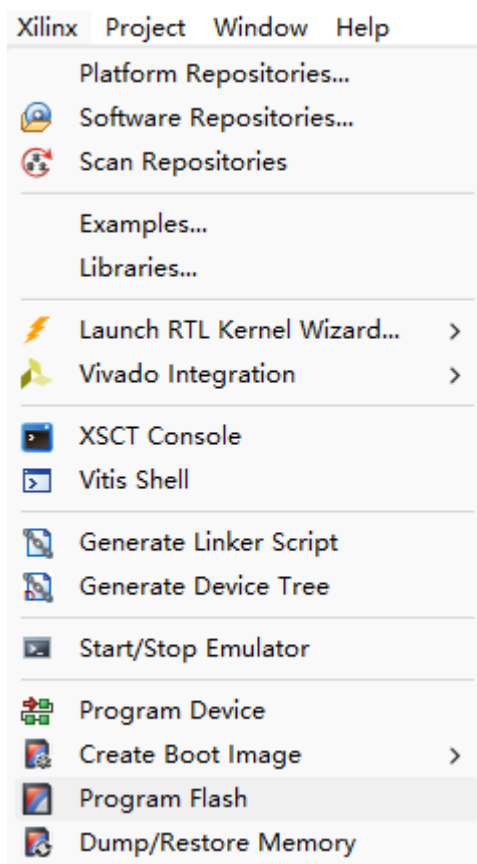
- 点击”Open target->Auto Connect”，等待连接。



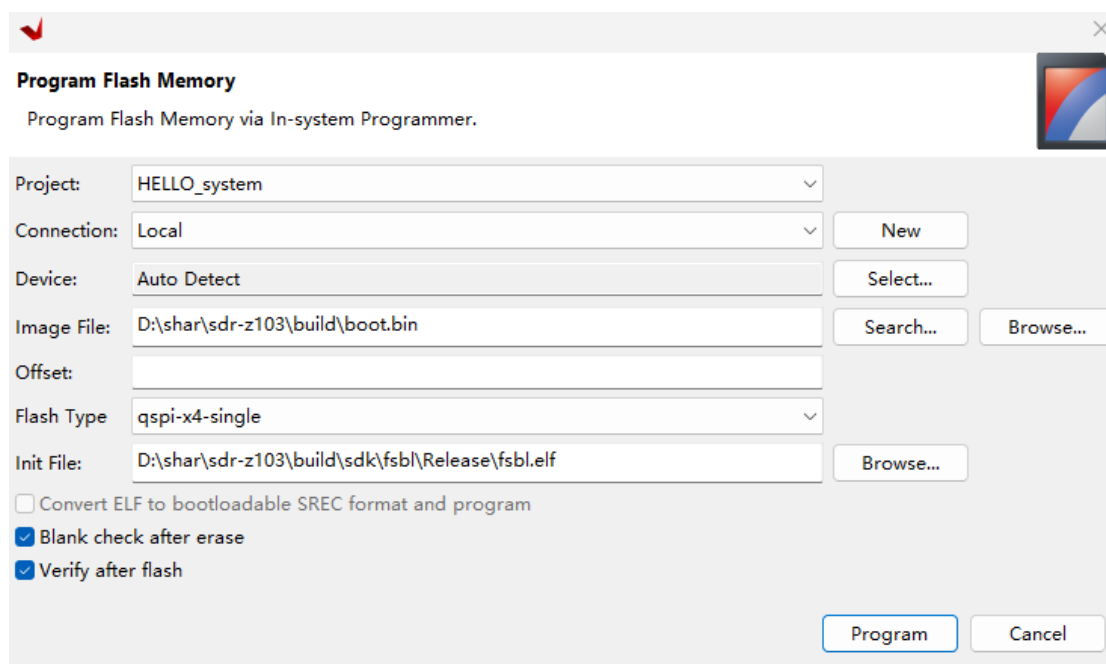
- 连接成功后，会看到 ARM 和 FPGA 在线状态正常，关闭 Vivado 软件即可。

| Name                              | Status       |
|-----------------------------------|--------------|
| localhost (1)                     | Connected    |
| xilinx_tcf/Digilent/EILWTCJIFTFR/ | Open         |
| arm_dap_0 (0)                     | N/A          |
| xc7z010_1 (0)                     | Not programr |

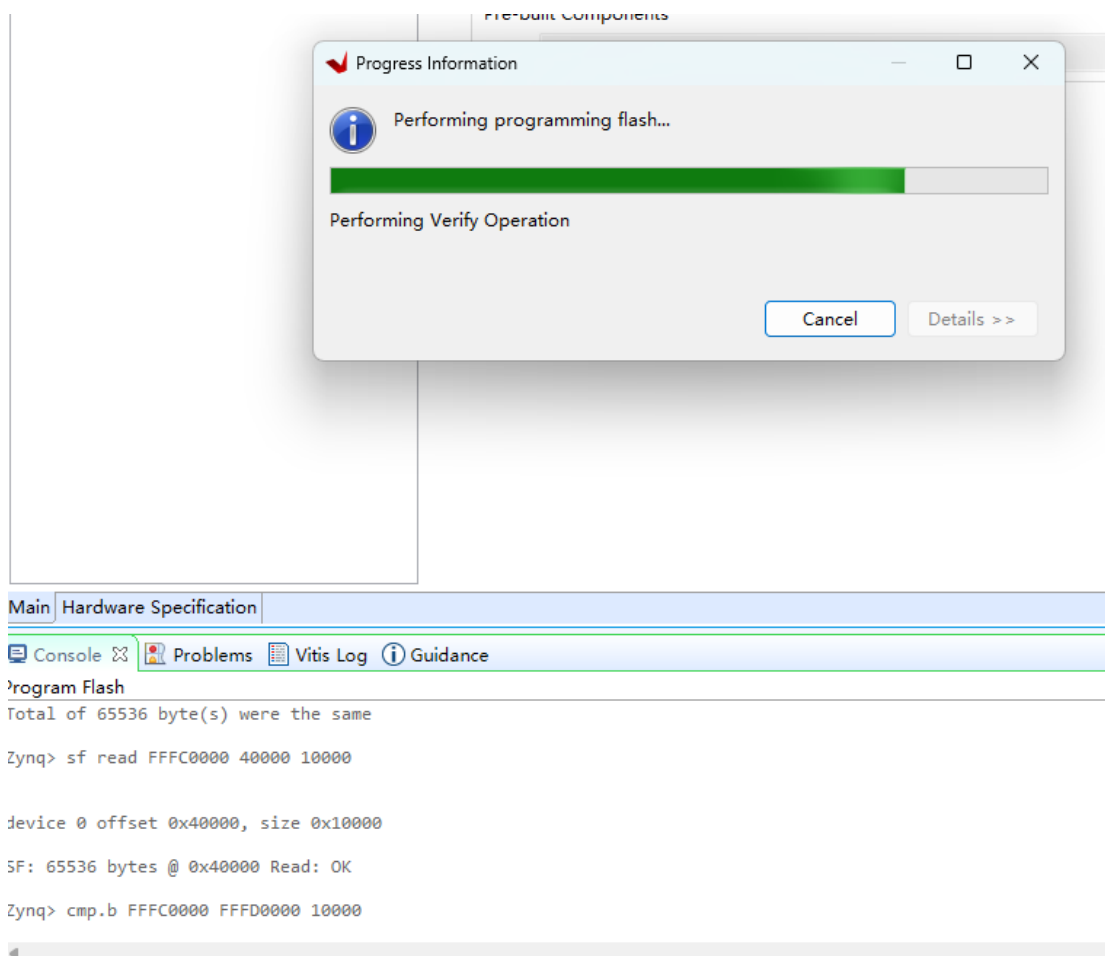
- 打开任意 Vitis/SDK 工程，点击菜单栏“Xilinx->Program Flash”。



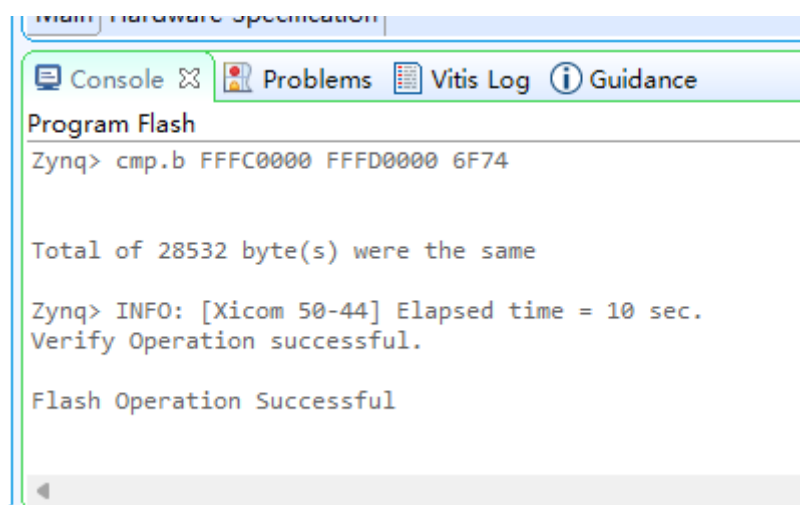
- 在 Image file 一栏点击 “Browse...”，选中出厂固件 boot.bin。在 Init File 一栏点击 “Browse...”，选中出厂固件 fsbl.elf，注意不能有中文路径。最后点击 “Program”。



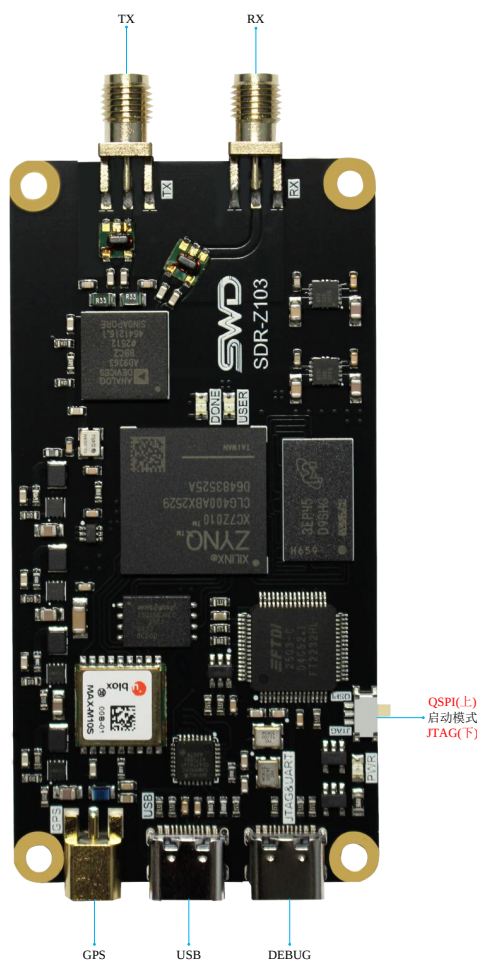
- 等待 Flash 固化完成。



- 固化成功后，会提示“Flash Operation Successful”信息，然后关闭 Vitis/SDK 即可。



- 板卡断电，将启动模式设置为“QSPI”。



- 按住 DFU 按键，将板卡 USB 口线缆插入到计算机 USB 口，DEBUG 口线缆可以不接，本文档例程也同时接入电脑 USB 口，查看烧录状态。当 USER 灯高亮，或“设备管理器->通用串行总线设备”中出现“USB download gadget”设备时，松开 DFU 按键即可。





- 将固件放在 C 盘，打开计算机命令提示符 CMD，切换到固件文件夹，注意不要有中文路径。
- 先烧录 Pluto 固件，输入 UPDATE.BAT pluto.dfu，Pluto 固件较大，烧录较慢，等待完成即可。

```
命令提示符
Microsoft Windows [版本 10.0.26100.7462]
(c) Microsoft Corporation. 保留所有权利。

C:\Users\76501>cd c:\build

c:\build>UPDATE.BAT pluto.dfu
dfu-util 0.9

Copyright 2005-2009 Weston Schmidt, Harald Welte and OpenMoko Inc.
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Please report bugs to http://sourceforge.net/p/dfu-util/tickets/

Opening DFU capable USB device...
ID 0456:b674
Run-time device DFU version 0110
Claiming USB DFU Interface...
Setting Alternate Setting #1 ...
Determining device status: state = dfuIDLE, status = 0
dfuIDLE, continuing
DFU mode device DFU version 0110
Device returned transfer size 4096
Copying data from PC to DFU device
Download [=====] 100% 12327371 bytes
Download done.
state(7) = dfuMANIFEST, status(0) = No error condition is present
state(2) = dfuIDLE, status(0) = No error condition is present
Done!
```

- 然后烧录 UBOOT 固件，输入 UPDATE.BAT uboot-env.dfu。

```
c:\build>UPDATE.BAT uboot-env.dfu
dfu-util 0.9

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Opening DFU capable USB device...
ID 0456:b674
Run-time device DFU version 0110
Claiming USB DFU Interface...
Setting Alternate Setting #3 ...
Determining device status: state = dfuIDLE, status = 0
dfuIDLE, continuing
DFU mode device DFU version 0110
Device returned transfer size 4096
Copying data from PC to DFU device
Download [=====] 100% 131072 bytes
Download done.
state(7) = dfuMANIFEST, status(0) = No error condition is present
state(2) = dfuIDLE, status(0) = No error condition is present
Done!
```

- 串口同时会打印烧录进程和状态。

```
User sessions
COM46 USB Serial Port (COM46)
=====
dfu-util 0.9
Copyright 2005-2009 Weston Schmidt, Harald Welte and OpenMoko Inc.
Copyright 2010-2016 Tormod Volden and Stefan Schmidt
This program is Free Software and has ABSOLUTELY NO WARRANTY
Please report bugs to http://sourceforge.net/p/dfu-util/tickets/
Opening DFU capable USB device...
ID 0456:b674
Run-time device DFU version 0110
Claiming USB DFU Interface...
Setting Alternate Setting #3 ...
Determining device status: state = dfuIDLE, status = 0
dfuIDLE, continuing
DFU mode device DFU version 0110
Device returned transfer size 4096
Copying data from PC to DFU device
Download [=====] 100% 131072 bytes
Download done.
state(7) = dfuMANIFEST, status(0) = No error condition is present
state(2) = dfuIDLE, status(0) = No error condition is present
Done!
CtrlC to exit ...
CtrlC to exit ...
```

- 烧录完成后，断电重启板卡即可使用。